

# Disproof of Conjecture 00000003411

## Entropy rates are not globally Lipschitz

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The conjecture asserts Lipschitz dependence of stationary Markov-chain entropy rates on transition probabilities, with a constant described in terms of minimal distribution values. There is no global Lipschitz dependence on the stated class. A uniform positive lower bound on the probabilities is not among the hypotheses.

Consider the two-state chain with transition matrix and stationary distribution

$$P_p = \begin{pmatrix} p & 1-p \\ p & 1-p \end{pmatrix}, \quad \pi_p = (p, 1-p), \quad 0 \leq p \leq 1.$$

Both rows have nonnegative entries summing to one, and  $\pi_p P_p = \pi_p$ . The chain started in this distribution is stationary; its successive states are independent with common distribution  $\pi_p$ . The usual entropy rate is

$$h(P_p, \pi_p) = - \sum_i \pi_p(i) \sum_j P_p(i, j) \log P_p(i, j) = H(p) := -p \log p - (1-p) \log(1-p).$$

As usual,  $0 \log 0 = 0$ , so  $H(0) = 0$ . Using natural logarithms changes only a fixed factor compared with any other logarithm base. In the maximum-entry norm on transition matrices,

$$\|P_p - P_q\|_\infty = \max_{i,j} |P_p(i, j) - P_q(i, j)| = |p - q|.$$

Thus the scalar parameter distance here is exactly the distance of the actual transition matrices.

**Theorem 1.** *No finite constant  $C \geq 0$  satisfies  $|h(P, \pi) - h(Q, \rho)| \leq C \|P - Q\|_\infty$  for all stationary finite-state Markov chains, even with two states.*

*Proof.* Given  $C \geq 0$ , choose  $p = e^{-(C+1)}$ , so  $0 < p < 1$  and  $\log p = -(C+1)$ . Since  $\log(1-p) \leq 0$ ,

$$H(p) \geq -p \log p = (C+1)p > Cp.$$

But  $\|P_p - P_0\|_\infty = p$  and  $H(0) = 0$ . This contradicts the proposed bound.  $\square$

All norms on the finite-dimensional space of two-by-two matrices are equivalent, so replacing the maximum-entry norm by another fixed matrix norm does not restore a global Lipschitz bound. Continuity itself is consistent with this example; the stronger claimed Lipschitz property fails. An estimate restricted to a class whose positive entries are uniformly bounded away from zero would be a different statement.

**Formal verification.** The Lean project defines the actual two-state distribution, transition kernel, row-stochastic and stationarity conditions, and the standard finite-sum entropy-rate functional. It proves validity, evaluates that functional, proves the exact maximum-entry transition-distance identity using the standard function-space metric, and constructs the explicit counterexample for every  $C$ . The final theorem negates a global Lipschitz bound even on this valid family. The definitions and proof do not assume an entropy or transition-distance certificate. Final axiom audits use only standard Lean axioms.